

Currently, Vietnam is increasingly emerging as one of the countries exporting information technology to the world. With a high-quality workforce, if equipped with global knowledge and skills, Vietnam can seize the opportunities of the ongoing information technology revolution.



Software Engineering is a field of study that focuses on applying engineering principles to design, develop, test, and maintain software systems efficiently and professionally. The core difference between Software Engineering and Computer Science lies in the training objectives. While Computer Science focuses on theoretical principles, algorithms, and scientific foundations, Software Engineering is oriented toward “building” real-world software products.

Mobile applications such as Grab, Momo, Zalo, or e-commerce platforms such as Shopee and Tiki are all the results of professional software development processes. Not only that, enterprise management systems (ERP), accounting software, and online banking systems are all created by software engineering experts.

The development of Industry 4.0, artificial intelligence, the Internet of Things (IoT), and cloud computing has created a large demand for software engineers. They need to have the ability to develop advanced technological solutions, contributing to the development of the digital economy.

Essential Qualities for Software Engineering Students

Passion for technology

In information technology, passion is an important factor that helps you easily adapt and integrate into the technology world. With adaptability, you will overcome the pressure and stress of work. Just think about it, without passion, could you spend hours sitting at a computer writing software, or even months finding a solution to a problem?

In addition, passion for technology will always inspire you and motivate you to invest in your products. Curiosity about technology and the desire to understand it will help you not feel bored while doing your work.

Creativity

In reality, intelligent and creative people tend to succeed more easily in the field of information technology. This profession always requires strong analytical thinking and the ability to optimize solutions to save cost and time while achieving the highest efficiency. Creativity and continuous practice of creativity will help you come up with highly applicable products that make users' lives easier. Your creativity in IT reflects your understanding of life and the needs of both yourself and society. This is the mission of Software Engineering and also the mission of professionals in this field.

Accuracy and attention to detail in work

Accuracy is a mandatory requirement in technology, including computer science. When building an application or software, just one small mistake can cause the entire program to fail. Precision and attention to detail in every aspect and every line of code are essential. This helps you save time and avoid spending too much time fixing errors when the software does not run properly. Your work quality and efficiency will also improve when you stay focused.

Good foreign language skills

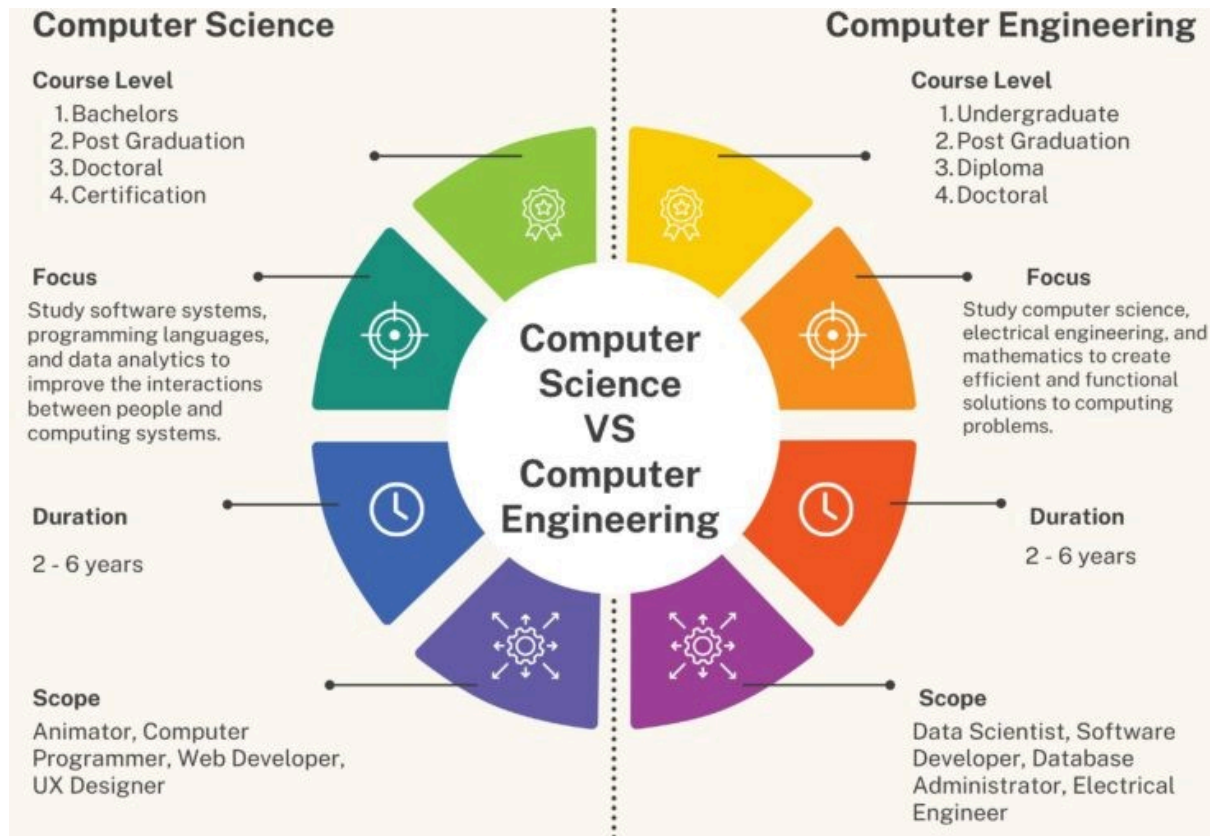
As a global system, becoming a good IT professional requires proficiency in foreign languages. You need English to read information, technical terms, and understand them accurately. The IT industry evolves every day worldwide, so you need language skills to keep up with global trends. Major technology companies in Vietnam such as FPT or Viettel are also expanding internationally. Mastering a global language will give you confidence and more career opportunities.

Eagerness to learn and improve knowledge

The world is constantly changing and developing, and today's knowledge can quickly become outdated. Therefore, you must continuously seek information and improve your knowledge in technology to keep up with development from both real-world

environments and training. Regularly updating knowledge will prevent you from falling behind in this rapidly evolving digital era. Learning and improving also help you accumulate more knowledge over time. Gradually, you will build a strong knowledge base that everyone desires.

Comparison between Software Engineering and Computer Science



What do Software Engineering students learn?

The Software Engineering curriculum is usually divided into main knowledge blocks, each playing a role in forming a complete software engineer. The logical arrangement of subjects helps students easily absorb and apply knowledge in practice.

Foundational and core knowledge block

This group of subjects helps students develop logical thinking, problem-solving abilities, and access basic technologies in the software field. These are the foundations for learning advanced subjects and applying them in real work later. Typical subjects include:

- Introduction to Programming
- Data Structures and Algorithms
- Databases
- Computer Networks

Specialized knowledge block

This group equips students with practical skills to participate directly in professional software development processes. It is an important stepping stone for students to become familiar with modern technologies and real business requirements. Some notable subjects include:

- Software Analysis and Design
- Software Testing (QA/QC)
- Software Project Management
- Web and Mobile Application Development

Career opportunities in Software Engineering

In the era of global digital transformation, Software Engineering in Vietnam is witnessing a strong boom in workforce demand. According to the latest report from TopDev (2024), Vietnam's IT labor market needs more than 600,000 additional workers by 2025, with software engineering accounting for the largest proportion.

Graduates in this field not only have opportunities to work in domestic companies but can also access the international job market thanks to global skills and strong expertise.

Typical job positions include:

- Software Developer
- Software Engineer
- Software Tester / QA Engineer
- Mobile/Web Application Developer
- DevOps / Cloud Engineer
- Data Engineer
- AI/ML Engineer or R&D Specialist
- Tech Lead, Software Project Manager

Along with the expansion of major technology companies such as FPT, Viettel, VNPT, VNG, and international companies in Vietnam such as Intel, Samsung, Bosch, KPMG,

Grab, Shopee..., opportunities for Software Engineering students are increasingly diverse and abundant.

In addition, the trend of remote work is growing rapidly, allowing Vietnamese software engineers to work directly for companies in the US, Australia, Singapore, or Europe with attractive salaries, opening global career opportunities while staying in Vietnam.

Salary in Software Engineering today

Software Engineering is currently among the fields with the highest and most stable income in Vietnam. According to TopDev 2024 salary reports:

- Fresh graduates: from 9 – 14 million VND/month
- Developers with 2–3 years of experience: from 18 – 30 million VND/month
- Senior Software Engineers: from 35 – 60 million VND/month
- Tech Lead / Project Manager: up to 70 – 100 million VND/month, especially in global tech companies

Income varies depending on job position, programming languages, workplace (Hanoi, Ho Chi Minh City, or remote), and company type (startup, local company, or foreign company).

Notably, the remote work trend with international companies allows Vietnamese software engineers to earn salaries in USD ranging from \$1,500 to \$3,000 per month.